

Final Exam D: Optimization, Lagrangians, and Dynamic Equations

GSE 510

Total Points: 100

10 Questions, 10 Points Each

Instructions: Answer all questions. Show your work clearly. Each question is worth 10 points. Partial credit will be given for correct reasoning.

1. Unconstrained Optimization: Local Maximum (10 points)

Consider the function

$$f(x, y) = 36x + 20y - 3x^2 - 5y^2.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Use the Hessian to classify the critical point.
- (e) Find the value of the function at the critical point.

2. Unconstrained Optimization: Saddle Point (10 points)

Consider

$$f(x, y) = -x^2 + 4y^2 + 10x - 16y.$$

- (a) Compute f_x and f_y .
- (b) Find the critical point.
- (c) Compute the Hessian matrix.
- (d) Compute the determinant of the Hessian.
- (e) Classify the critical point and explain briefly.

3. Profit Maximization with Two Choice Variables (10 points)

A firm chooses labor L and capital K . Profit is

$$\pi(L, K) = 100L + 60K - 4L^2 - 2K^2 - 4LK.$$

- (a) Compute π_L and π_K .
- (b) Solve the first-order conditions.
- (c) Compute the Hessian matrix.
- (d) Determine whether the critical point is a local maximum.

- (e) Interpret the cross-partial derivative π_{LK} .

4. Lagrangian: Consumer Choice

(10 points)

Consider the problem

$$\max_{x,y} x^{2/5} y^{3/5}$$

subject to

$$4x + 3y = 60.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions.
- (c) Derive the relationship between x and y at an interior optimum.
- (d) Solve for x^* and y^* .
- (e) Interpret the Lagrange multiplier in words.

5. Bordered Hessian

(10 points)

Consider

$$\max_{x,y} xy$$

subject to

$$x + 3y = 24.$$

- (a) Write the Lagrangian.
- (b) Solve the first-order conditions.
- (c) Compute the bordered Hessian.
- (d) Evaluate the bordered Hessian at the critical point.
- (e) Use the bordered Hessian condition discussed in class to classify the critical point.

6. When the Lagrangian Is Not the Whole Story

(10 points)

A consumer has utility

$$u(x, y) = 2x + 5y$$

and budget constraint

$$x + y = 10,$$

with

$$x \geq 0, \quad y \geq 0.$$

- (a) Write the Lagrangian for the equality-constrained problem.
- (b) Compute the first-order conditions.
- (c) Explain why the standard interior Lagrangian method does not deliver an interior solution.
- (d) Evaluate utility at $(10, 0)$ and $(0, 10)$.

- (e) Find the true utility-maximizing bundle.
- (f) Explain why this is a corner solution.

7. Constrained Optimization with Boundary Check

(10 points)

Consider

$$\max_{x,y} x^3 y$$

subject to

$$x + y = 20,$$

with

$$x \geq 0, \quad y \geq 0.$$

- (a) Write the Lagrangian.
- (b) Compute the first-order conditions for an interior solution.
- (c) Solve for the interior critical point.
- (d) Evaluate the objective at the interior critical point.
- (e) Evaluate the objective at the endpoints $(20, 0)$ and $(0, 20)$.
- (f) Is the interior critical point the global maximum?

8. Linear Difference Equation

(10 points)

Consider

$$x_{t+1} = 0.8x_t + 6.$$

- (a) Find the steady state x^* .
- (b) Suppose $x_0 = 0$. Compute x_1 , x_2 , and x_3 .
- (c) Write the equation in the form

$$x_{t+1} - x^* = a(x_t - x^*).$$

- (d) Is the steady state stable?
- (e) Explain the economic meaning if x_t is capital.

9. Differential Equation and Stability

(10 points)

Consider

$$\frac{dp}{dt} = 0.25(D(p) - S(p)),$$

where

$$D(p) = 200 - 4p, \quad S(p) = 50 + p.$$

- (a) Substitute demand and supply into the differential equation.
- (b) Simplify the differential equation.

- (c) Find the steady-state price.
- (d) Determine whether the steady state is stable.
- (e) Interpret the adjustment rule economically.

10. Solow-Style Difference Equation

(10 points)

Consider the capital accumulation equation

$$k_{t+1} = sAk_t^{1/2} + (1 - \delta)k_t,$$

where

$$s = 0.3, \quad A = 10, \quad \delta = 0.2.$$

- (a) Substitute the parameter values into the equation.
- (b) Find the steady state k^* .
- (c) Suppose $k_0 = 100$. Compute k_1 .
- (d) Suppose $k_0 = 400$. Compute k_1 .
- (e) Based on parts (c) and (d), explain whether capital moves toward the steady state.
- (f) Interpret the steady state economically.